



Comparative Study of Homogeneous and Sampling Electromagnetic Calorimeters

A Geant4 and ROOT Analysis of Shower Containment, Passive Material Effects, and Sampling Fluctuations in $PbWO_4$ and LAr/Pb

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Introduction to the simulated calorimeters

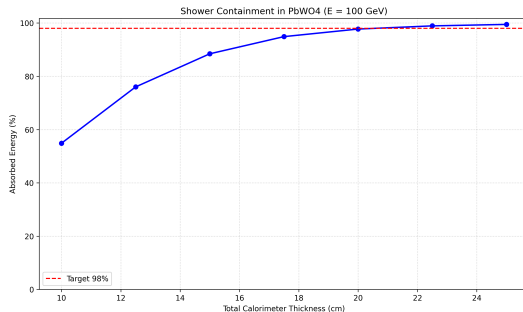
	Homogeneous (CMS)	Sampling (ATLAS)
Material	Lead Tungstate ($PbWO_4$) crystals	Lead (Pb) absorber + Liquid Argon (LAr) active medium
Working Principle	A single medium acts simultaneously as both the full absorber and the active detector.	Passive layers absorb the majority of energy, while active layers sample the shower.
Characteristics	Short X_0 (≈ 0.89 cm) results in a highly compact design with good energy resolution.	Highly radiation-resistant design for optimal longitudinal segmentation.



Electromagnetic shower containment in PbWO_4

Simulating 100 GeV electrons:

- Lead Tungstate (PbWO_4) is extremely dense, featuring a short radiation length ($X_0 \approx 0.89$ cm).
- Standard electromagnetic calorimeter design requires approximately $25 X_0$ to effectively contain the shower and its tail.
- The 98% energy containment threshold to be reached at approximately 22 cm of thickness.

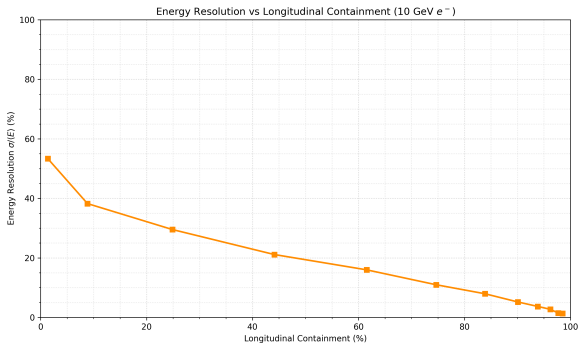


The simulation confirms the expected longitudinal development.



Impact of longitudinal leakage on energy resolution

Incomplete longitudinal containment introduces severe statistical fluctuations in the measured energy. As the containment approaches 100%, leakage vanishes and the energy resolution $\sigma/\langle E \rangle$ drops, as demonstrated below for 10 GeV e^- .

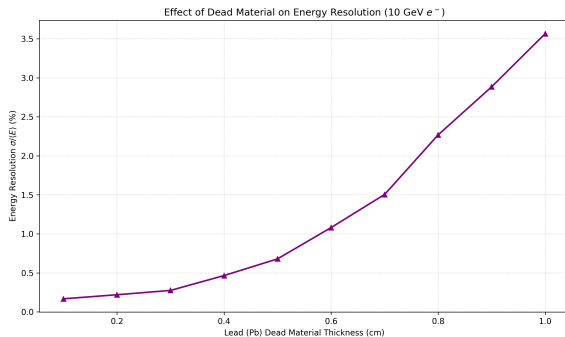


Depth [cm]	Depth [X_0]	Cont. [%]	$\sigma/\langle E \rangle$ [%]
2.50	2.81	1.35	53.35
4.00	4.49	8.84	38.25
5.50	6.18	24.89	29.52
8.50	9.55	61.54	15.99
10.00	11.24	74.65	10.99
11.50	12.92	83.88	7.95
13.00	14.61	90.09	5.19
14.50	16.29	93.86	3.69
19.00	21.35	98.53	1.31



Effect of upstream dead material on resolution

The presence of inactive material (e.g., a tracker) in front of the calorimeter negatively impacts energy resolution. Simulating 10 GeV e^- traversing a Lead (Pb) layer from 0.1 to 1.0 cm demonstrates how unmeasured energy loss from early shower initiation causes a severe degradation in performance.



Pb Depth [cm]	$\sigma/\langle E \rangle$ [%]
0.10	0.17
0.20	0.22
0.30	0.27
0.80	2.27
0.90	2.89
1.00	3.57

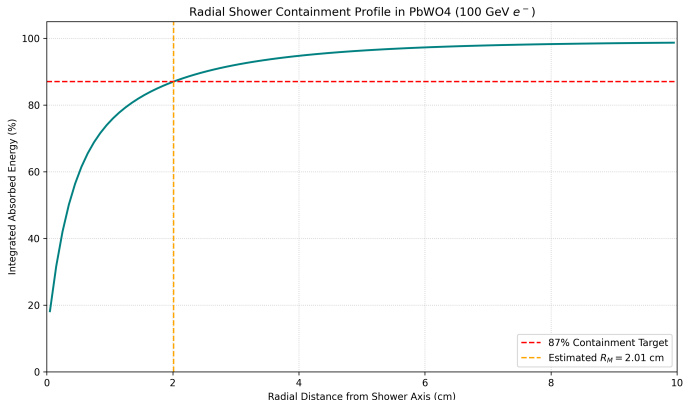


Optimal cell size: transverse containment

The ideal cell size is dictated by the Molière radius (R_M), which encloses $\sim 87\%$ of the transverse shower energy.

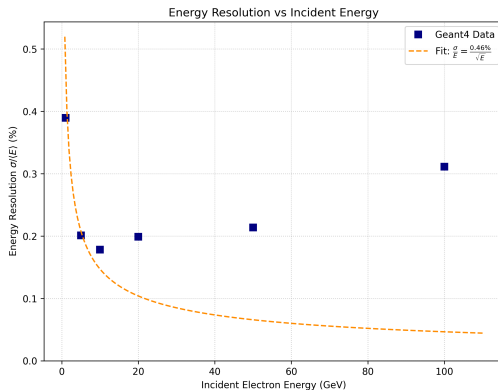
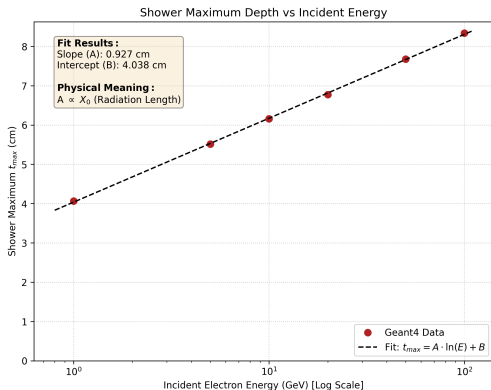
- $R_M(\text{PbWO}_4) = 2.2 \text{ cm}$

Simulation reaches the 87% target exactly at $R = 2.01 \text{ cm}$.





Energy response: the resolution anomaly





Understanding the resolution anomaly

- As predicted by cascade theory, the shower maximum (t_{max}) strictly scales with $\ln(E)$.
- Since the Geant4 simulation does not model photon production and collection inefficiencies, the intrinsic stochastic term is artificially tiny ($\sim 0.46\%/\sqrt{E}$).
- Because this baseline intrinsic error is so close to zero, even minor event-by-event fluctuations caused by longitudinal leakage severely dominate the total resolution. This entirely breaks the ideal $1/\sqrt{E}$ scaling at high energies.
- This effect is dramatically visible here, but it is usually masked in sampling calorimeters. In those setups, as we will see later, the inherently large sampling fluctuations ($\approx 10\%/\sqrt{E}$) easily cover up these small leakage penalties.



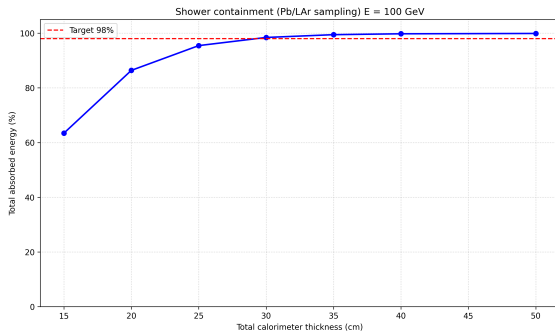
Sampling calorimeter dimensioning (Pb/LAr)

Simulating 100 GeV electrons.

The setup uses 50 layers with a 3:4 thickness ratio for Pb and LAr respectively.

- $X_0(\text{Pb}) = 0.56 \text{ cm}$
- $X_0(\text{LAr}) = 14.0 \text{ cm}$

$$X_{\text{eff}} = 1.24 \text{ cm} \quad \text{Depth} \approx 25X_{\text{eff}} = 31 \text{ cm}$$

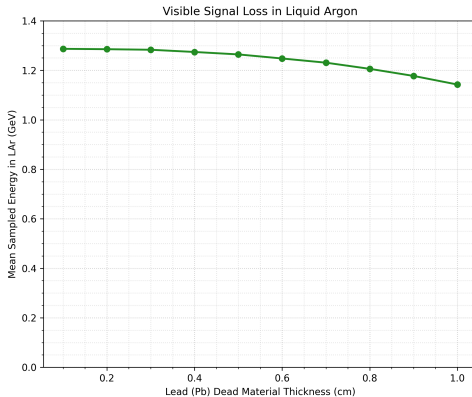
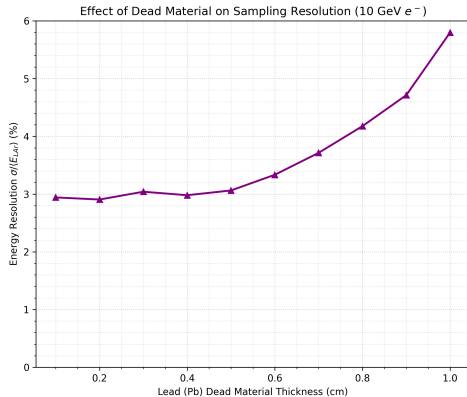


The simulation confirms the 98% absorbed energy threshold is successfully reached at 30 cm of total thickness.



Effect of upstream dead material (sampling)

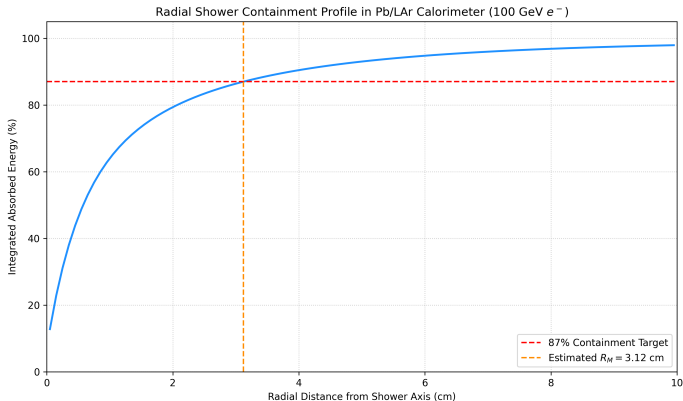
The unmeasured energy absorbed by the dead material fluctuates event-by-event, severely degrading the sampling resolution.





Optimal cell size: transverse containment (Pb/LAr)

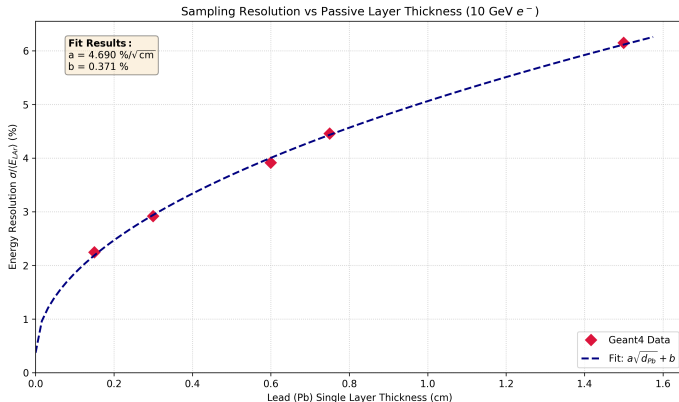
- Cell size is determined by the 87% transverse energy containment limit.
- The data shows an effective radius of $R_M = 3.12$ cm.





Sampling frequency & energy resolution

Fixed total material (15 cm Pb + 20 cm LAr). Layers varied from 100 down to 10. Thicker individual Pb absorbers reduce sampling frequency, worsening the energy resolution.

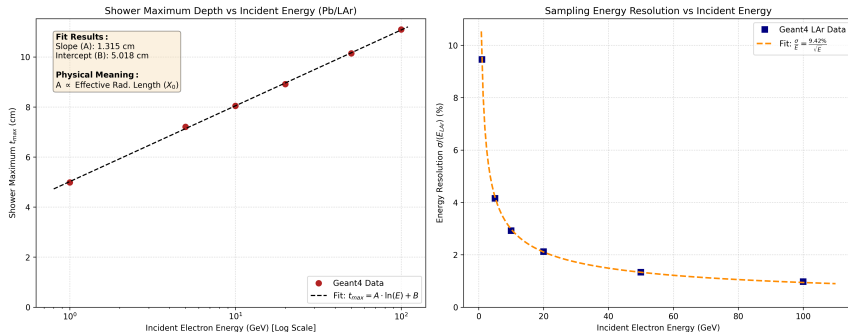




Sampling calorimeter response (1–100 GeV)

As expected from cascade models, the penetration depth t_{max} strictly follows a logarithmic scaling with energy $\ln(E)$.

The energy resolution shows an excellent agreement with the stochastic term $\sigma/E = a/\sqrt{E}$.

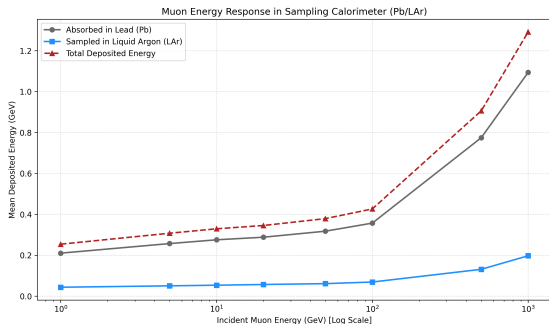




Muon energy response (Pb/LAr)

Up to ~ 100 GeV, muons act as Minimum Ionizing Particles (MIPs), depositing a small and nearly constant fraction of their energy via ionization.

At ultra-high energies (500–1000 GeV), radiative losses (e.g., bremsstrahlung) become dominant, causing a sharp increase in the deposited energy.





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Thank you for listening!
Any questions?

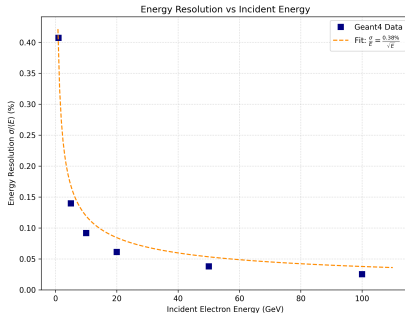


BACKUP



Resolving the resolution anomaly

In a pure Geant4 simulation lacking photon collection inefficiencies and electronic noise, the intrinsic stochastic term is extremely small. Therefore, even minor event-by-event fluctuations caused by longitudinal leakage become the dominant source of error at high energies, breaking the ideal $\propto 1/\sqrt{E}$ scaling. By doubling the $PbWO_4$ detector length to 50 cm, the shower tail is fully contained.





Fluctuations in sampling calorimeters

According to Rossi's Approximation B, the energy resolution is dominated by fluctuations in the number of charged particle crossings (N_x) in the active layers.

- **Visible energy (E_{vis}):**

$$E_{vis} = N_x \Delta E_{active} = N_x \left(\frac{dE}{dx} \right)_{active} d_{active}$$

- **Number of crossings (N_x):** Expressed via the sampling fraction f_{samp} :

$$f_{samp} = \frac{E_{vis}}{E} \implies N_x = \frac{f_{samp} E}{\Delta E_{active}}$$

- **Poisson fluctuations:**

$$\frac{\sigma(E_{vis})}{E_{vis}} = \frac{\sigma(N_x)}{N_x} = \frac{1}{\sqrt{N_x}} = \sqrt{\frac{\Delta E_{active}}{f_{samp} E}} \propto \sqrt{\frac{d_{active}}{f_{samp}}} \frac{1}{\sqrt{E}}$$

At a fixed sampling fraction f_{samp} , the ratio of energy deposited in the active vs. passive layers is constant. Thus, the thicknesses are strictly proportional: $d_{active} \propto C d_{passive}$.